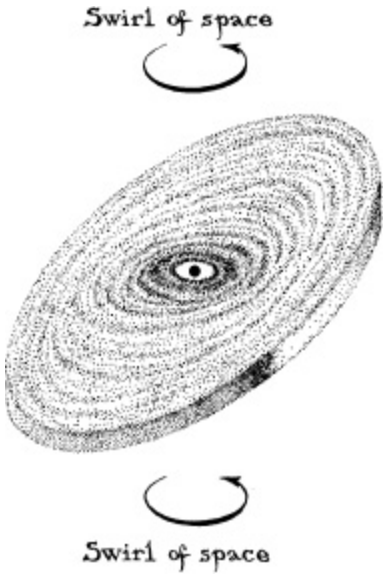
Without Kerr's solution to the Einstein field equation, this gyroscopic action would have been unknown, and it might have been impossible to explain quasars. With Kerr's solution in hand, astrophysicists in the mid-1970s were arriving at a clear and elegant explanation. For the first time, the concept of a black hole as a dynamical body, more than just a "hole in space," was playing a central role in explaining astronomers' observations.

How strong will the swirl of space be near the gigantic hole? In other words, how fast will gigantic holes spin? James Bardeen deduced the answer: He showed mathematically that gas accreting into the hole from its disk should gradually make the hole spin faster and faster. By the time the hole has swallowed enough inspiraling gas to double its mass, the hole should be spinning at nearly its maximum possible rate—the rate beyond which centrifugal forces prevent any further speedup (Chapter 7). Thus, gigantic holes should typically have near-maximal spins.



9.6 The spin of a black hole produces a swirl of space around the hole, and that swirl holds the inner part of the accretion disk in the hole's equatorial plane.

How can a black hole and its disk produce two oppositely pointed jets? Amazingly easily, Blandford, Rees, and Lynden-Bell at Cambridge University recognized in the mid-1970s. There are four possible ways to produce jets; anyone of them might do the job.

First, Blandford and Rees realized, the disk may be surrounded by a cool gas cloud (Figure 9.7a). A wind blowing off the upper and lower faces